Analyzing Transformations with Multiple Parameters Answer Key

Precalculus

Quick Answers

1. 9
2. (a) g(-2) = -3, —
3. $\frac{(-1)x^2}{2} - 4x - 11$
4. (a) g(1) = 2, (b) g(2) = 14
5. (a) g(9) = -7, —
6. $x^2 - 8x + 17$
7. (a) q(0) = 3, (b) q(3) = -6
8. (a) g(2) = 64, —
9. $3\sqrt{2\,(x-4)} + 1$
10. (a) vertex form = $4\,(x+3)^2 + 5$, —

What is verified

12 of 16 answers machine-checked against SymPy, across 10 problems.
— marks an answer only you can judge — problems 2, 5, 8, 10. The worked solution states what a correct response must contain.

Curriculum

Level: Precalculus
HSF-BF.B.3 – *problems 2, 3, 5, 6, 8, 9, 10*
HSF-IF.A – *problems 1, 4, 7*
Difficulty 1–5 of 5 across 16 tagged checks.

1. $f(x) = x^2$, $g(x) = 2f(x-3) + 1$. Find $g(5)$.

Solution: Work inside first. $g(5) = 2f(5-3)+1 = 2f(2)+1 = 2(4)+1 = 9$. $\boxed{g(5) = 9}$

2. $f(x) = x^2$, $g(x) = f(x+4) - 7$. **(a)** Find $g(-2)$. **(b)** Describe the transformations in order.

Solution (a): $g(-2) = f(-2+4) - 7 = f(2) - 7 = 4 - 7 = -3$. $\boxed{g(-2) = -3}$

Solution (b): *Instructor-judged.* A complete answer names two shifts with the right directions: a horizontal shift of 4 units **left** (the +4 is inside, so the graph moves toward negative x — opposite to the sign's appearance), then a vertical shift of 7 units **down**. Because one shift is horizontal and one is vertical, either order gives the same graph, and a student may say so. Reading $x + 4$ as "4 right" is the standard error and should not earn that half.

3. $f(x) = x^2$, $g(x) = -\dfrac{1}{2}f(x+4) - 3$. Expand.

Solution: Substitute, expand the square, then distribute the $-\frac{1}{2}$.

$$\begin{aligned} g(x) &= -\tfrac{1}{2}(x+4)^2 - 3 \\ &= -\tfrac{1}{2}\left(x^2 + 8x + 16\right) - 3 \\ &= -\tfrac{1}{2}x^2 - 4x - 8 - 3 \end{aligned}$$

So $\boxed{g(x) = -\frac{1}{2}x^2 - 4x - 11}$

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4. $f(x) = x^2 + 3$, $g(x) = f(2x) - 5$. **(a)** Find $g(1)$. **(b)** Find $g(2)$.

Solution (a): The $2x$ doubles the input before f sees it: $g(1) = f(2) - 5$. The table gives $f(2) = 7$, so $g(1) = 7 - 5 = 2$. $\boxed{g(1) = 2}$

Solution (b): $g(2) = f(4) - 5$, and 4 is off the end of the table, so use the rule: $f(4) = 4^2 + 3 = 19$, hence $g(2) = 19 - 5 = 14$. $\boxed{g(2) = 14}$

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5. $f(x) = \sqrt{x}$, $g(x) = -3f(x) + 2$. **(a)** Find $g(9)$. **(b)** Describe the transformations in order.

Solution (a): $g(9) = -3\sqrt{9} + 2 = -3(3) + 2 = -7$. $\boxed{g(9) = -7}$

Solution (b): *Instructor-judged.* Every parameter is outside f, so nothing at all happens horizontally. A complete answer gives a vertical stretch by a factor of 3 and a reflection in the x-axis (these two may be combined as "multiply the outputs by -3"), **then** a shift of 2 units up. The stretch must come before the vertical shift; a horizontal stretch or a left/right shift is wrong.

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6. $f(x) = x^2 - 4x$, $g(x) = f(x-2) + 5$. Expand.

Solution: Replace every x in the rule for f by $x - 2$:

$$\begin{aligned} g(x) &= (x-2)^2 - 4(x-2) + 5 \\ &= x^2 - 4x + 4 - 4x + 8 + 5 \end{aligned}$$

Collecting like terms gives $\boxed{g(x) = x^2 - 8x + 17}$

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7. $p(x) = x^3$, $q(x) = -p(x-1) + 2$. **(a)** Find $q(0)$. **(b)** Find $q(3)$.

Solution (a): $q(0) = -p(0-1) + 2 = -p(-1) + 2 = -(-1) + 2 = 3$. $\boxed{q(0) = 3}$

Solution (b): $q(3) = -p(3-1) + 2 = -p(2) + 2 = -8 + 2 = -6$. $\boxed{q(3) = -6}$

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8. $f(x) = x^3$, $g(x) = f(6 - x)$. **(a)** Find $g(2)$. **(b)** Rewrite the input and describe the transformations in order.

Solution (a): $g(2) = f(6-2) = f(4) = 4^3 = 64$. $\boxed{g(2) = 64}$

Solution (b): *Instructor-judged.* The input factors as $6 - x = -(x - 6)$, and until it is written that way no transformation can be read off. A complete answer then names a reflection in the y-axis and a horizontal shift of 6 units **right**, with an order that actually works: reflect in the y-axis first and then shift right 6. Reading $6 - x$ unfactored and answering "shift 6 left" is the error this problem targets.

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9. $f(x) = \sqrt{x}$, $g(x) = 3f(2x - 8) + 1$. Rewrite as $a\sqrt{b(x-h)} + k$.

Solution: The parameters are only readable once b is factored out of the input:

$$2x - 8 = 2(x - 4)$$

So $g(x) = 3\sqrt{2(x-4)} + 1$: a horizontal compression by a factor of $\frac{1}{2}$, a shift 4 right, a vertical stretch by 3, and a shift 1 up. $\boxed{g(x) = 3\sqrt{2(x-4)} + 1}$

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10. $g(x) = 4x^2 + 24x + 41$. **(a)** Complete the square. **(b)** Describe the transformations in order.

Solution (a): Factor the leading coefficient out of the x-terms first:

$$\begin{aligned} g(x) &= 4\left(x^2 + 6x\right) + 41 \\ &= 4\left(x^2 + 6x + 9\right) - 36 + 41 \\ &= 4(x+3)^2 + 5 \end{aligned}$$

So $\boxed{g(x) = 4(x+3)^2 + 5}$

Solution (b): *Instructor-judged.* From $4(x+3)^2 + 5$: a vertical stretch by a factor of 4, a shift 3 units **left**, and a shift 5 units **up**. The order that matters is stretch-before-vertical-shift: applying the $+5$ first and then stretching by 4 would multiply that 5 as well and land the vertex at $y = 20$ instead of $y = 5$. The horizontal shift may be done at any point.